

Date: 2026-06-30
Request ID: 100048709

Hotplate		
Analyte(s)	Co, K, Pt	
SOP#		
Cocktail	15 mL HCl, 10 mL H2SO4, 5 mL HNO3	
sample(s)	weight (g)	volume (mL)
200128418_1	0.5074	50
200128418_2	0.5059	50

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LOI temp: 900 °C

sample	crucible (g)	crucible + sample (g)	after 900 °C	LOI (%)	correction
200128418	16.9723	18.9821	18.9142	3.38	0.9662

Co ICP analysis

sample	Dilution	conc. [mg/L]	Co_ppm [dried]	%Co [dried]
200128418_1	100/0.2	2.198	112083.89	11.21
200128418_2	100/0.18	1.948	110700.03	11.07

co result: 111391.96 ppm

co result: 11.14 %

co lot: *Manufacturer: Assurance, 29-22COY exp: 2027-06-30*

K ICP analysis

sample	Dilution	conc. [mg/L]	K_ppm [dried]	%K [dried]
200128418_1	100/0.8	0.972	12391.44	1.24
200128418_2	100/0.8	0.952	12166.06	1.22

k result: 12278.75 ppm

k result: 1.23 %

k lot: *Manufacturer: Assurance, 28-6KY exp: 2026-08-30*

Pt ICP analysis

sample	Dilution	conc. [mg/L]	Pt_ppm [dried]	%Pt [dried]
200128418_1	10/7	1.419	206.74	0.02
200128418_2	50/50	2.023	206.88	0.02

pt result: 206.81 ppm

pt result: 0.02 %

pt lot: *Manufacturer: Assurance, 28-2PTY exp: 2027-02-28*

Note(s):

$$\text{ppm } M+ = [\text{conc.}][\text{volume}][\text{dilution}]/[\text{weight}]$$

$$\%M+ = [\text{conc.}][\text{volume}][\text{dilution}]/([\text{weight}]*10,000)$$

$$\%MO = [\text{oxide factor}]*\%M+$$

A = crucible

B = crucible + sample

C = crucible + sample after drying

$$\%LOI = ([B-A]-[C-A])/(B-A)*100\%$$

$$\text{ppm } M+ \text{ [dried]} = \text{ppm } M+ / ([1-(\%LOI)/100])$$

$$\%M+ \text{ [dried]} = \%M+ / ([1-(\%LOI)/100])$$

$$\%MO \text{ [dried]} = \%MO / ([1-(\%LOI)/100])$$